



Spring 2021–2022 (Math 102: Calculus II)

First Mid-Term Exam: Suggested solutions

1. **(10 Points)** Find, if possible, the following limit: $\lim_{x \rightarrow 0^-} \left(\frac{1}{x} - \frac{1}{e^x - 1} \right)$.

Answer: The limit is indeterminate of the form $-\infty + \infty$. Using a common denominator, we get:

$\lim_{x \rightarrow 0^-} \left(\frac{1}{x} - \frac{1}{e^x - 1} \right) = \lim_{x \rightarrow 0^-} \frac{e^x - 1 - x}{xe^x - x}$, which has the form $\frac{0}{0}$. Hence, we may use l'Hospital's Rule to obtain from the last limit: $\lim_{x \rightarrow 0^-} \frac{e^x - 1}{xe^x + e^x - 1}$, which has also the form $\frac{0}{0}$. Using once again l'Hospital's Rule, the last limit becomes: $\lim_{x \rightarrow 0^-} \frac{e^x}{xe^x + 2e^x} = 1/2$.

2. **(15 Points)** Evaluate the following integral $\int e^x \sin(2x) dx$.

Answer: We integrate by parts with, for instance, $u = \sin(2x)$ and $dv = e^x dx \Rightarrow du = 2 \cos(2x) dx$ and $v = e^x$. Hence $I := \int e^x \sin(2x) dx = e^x \sin(2x) - 2 \int e^x \cos(2x) dx$. Use again an integration by parts with $u = \cos(2x)$ and $dv = e^x dx$. Then $du = -2 \sin(2x) dx$ and $v = e^x$.

Thus $I = \int e^x \sin(2x) dx = e^x \sin(2x) - 2 \left(e^x \cos(2x) + 2 \int e^x \sin(2x) dx \right) \Rightarrow I = e^x \sin(2x) - 2e^x \cos(2x) - 4I$. Solving for I , we get $I = \int e^x \sin(2x) dx = \frac{e^x \sin(2x) - 2e^x \cos(2x)}{5} + C$.

3. **(15 Points)** Find $\int \frac{1}{2\sqrt{x+3} + x + 5} dx$.

Answer: We use the rationalization technique.

Let $u = \sqrt{x+3} \Rightarrow u^2 = x+3$ and so $x = u^2 - 3$, $dx = 2u du$. Substitute to get

$$I := \int \frac{1}{2\sqrt{x+3} + x + 5} dx = \int \frac{1}{2u + u^2 - 3 + 5} 2u du = \int \frac{2u}{u^2 + 2u + 2} du = \int \frac{2u}{(u+1)^2 + 1} du.$$

Let $t = u + 1 \Rightarrow u = t - 1$ and $du = dt$. Therefore

$$\begin{aligned} I &= \int \frac{2u}{(u+1)^2 + 1} du = \int \frac{2(t-1)}{t^2 + 1} dt = \int \frac{2t}{t^2 + 1} dt - 2 \int \frac{1}{t^2 + 1} dt \\ &= \ln(t^2 + 1) - 2 \tan^{-1} t + C = \ln(u^2 + 2u + 2) - 2 \tan^{-1}(u + 1) + C \\ &= \ln(x + 5 + 2\sqrt{x+3}) - 2 \tan^{-1}(1 + \sqrt{x+3}) + C. \end{aligned}$$

4. **(20 Points)** Evaluate $\int \frac{(\sin^2 \theta - 2 \sin \theta + 2) \cos \theta}{\sin^4 \theta + 2 \sin^2 \theta} d\theta$.

Answer: Let $x = \sin \theta \Rightarrow dx = \cos \theta d\theta$. Substitute to obtain:

$$I := \int \frac{\sin^2 \theta - 2 \sin \theta + 2}{\sin^4 \theta + 2 \sin^2 \theta} \cos \theta d\theta = \int \frac{x^2 - 2x + 2}{x^4 + 2x^2} dx.$$

Note that the integrand is a proper rational function and hence we can use partial fractions.

$$\frac{x^2 - 2x + 2}{x^4 + 2x^2} = \frac{x^2 - 2x + 2}{x^2(x^2 + 2)} = \frac{A}{x} + \frac{B}{x^2} + \frac{Cx + D}{x^2 + 2}.$$

Multiply both sides by $x^2(x^2 + 2)$, we obtain: $x^2 - 2x + 2 = (A + C)x^3 + (D + B)x^2 + 2Ax + 2B$. Therefore, $A = -1, B = 1, C = 1$ and $D = 0$. Hence

$$I = \int \left(-\frac{1}{x} + \frac{1}{x^2} + \frac{x}{x^2 + 2} \right) dx = -\ln|x| - \frac{1}{x} + \frac{1}{2} \ln(2 + x^2) + K, \quad K \in \mathbb{R}.$$

Since $x = \sin \theta$, we get

$$I = -\ln|\sin \theta| - \frac{1}{\sin \theta} + \frac{1}{2} \ln(2 + \sin^2 \theta) + K, \quad K \in \mathbb{R}.$$

5. **(20 Points)** Evaluate the definite integral $\int_1^{2/\sqrt{3}} \frac{\sqrt{x^2 - 1}}{x^4} dx$.

Answer: Use the trigonometric substitution $x = \sec \theta$, where $\theta = \sec^{-1} x \in [0, \pi/2) \cup [\pi, 3\pi/2)$. Then $dx = \sec \theta \tan \theta d\theta$ and $\sqrt{x^2 - 1} = \sqrt{\sec^2 \theta - 1} = \tan \theta$. We also change the limit of integration as follows: $x = 1 \Rightarrow \theta = \sec^{-1} 1 = 0$ and $x = 2/\sqrt{3} \Rightarrow \theta = \sec^{-1}(2/\sqrt{3}) = \pi/6$. Substitute to get

$$\int_1^{2/\sqrt{3}} \frac{\sqrt{x^2 - 1}}{x^4} dx = \int_0^{\pi/6} \frac{\tan \theta}{\sec^4 \theta} \sec \theta \tan \theta d\theta = \int_0^{\pi/6} \frac{\tan^2 \theta}{\sec^3 \theta} d\theta = \int_0^{\pi/6} \sin^2 \theta \cos \theta d\theta.$$

Let $u = \sin \theta \Rightarrow du = \cos \theta d\theta$. Moreover, the limits of integration for u are: $\theta = 0 \Rightarrow u = \sin 0 = 0$ and $\theta = \pi/6 \Rightarrow u = \sin(\pi/6) = 1/2$. Substitute to get

$$\int_1^{2/\sqrt{3}} \frac{\sqrt{x^2 - 1}}{x^4} dx = \int_0^{\pi/6} \sin^2 \theta \cos \theta d\theta = \int_0^{1/2} u^2 du = \frac{1}{3} [u^3]_0^{1/2} = \frac{1}{24}.$$

6. **(20 Points)** Determine whether the given improper integral is convergent or divergent. If it is convergent, then find its value.

$$\int_1^e \frac{1}{x\sqrt{1 - (\ln x)^2}} dx.$$

Answer: Note that the integrand is continuous on $[1, e)$ and goes to infinity as $x \rightarrow e^-$. Therefore

$$\int_1^e \frac{1}{x\sqrt{1 - (\ln x)^2}} dx = \lim_{t \rightarrow e^-} \left(\int_1^t \frac{1}{x\sqrt{1 - (\ln x)^2}} dx \right).$$

Let $u = \ln x$ and so $du = \frac{1}{x} dx$. Moreover, if $x = 1$, then $u = \ln 1 = 0$. In turn, if $x = t$, then $u = \ln t$. Substitute to get

$$\int_1^t \frac{1}{x\sqrt{1 - (\ln x)^2}} dx = \int_0^{\ln t} \frac{1}{\sqrt{1 - u^2}} du = [\sin^{-1} u]_0^{\ln t} = \sin^{-1}(\ln t).$$

Hence

$$\int_1^e \frac{1}{x\sqrt{1 - (\ln x)^2}} dx = \lim_{t \rightarrow e^-} \left(\int_1^t \frac{1}{x\sqrt{1 - (\ln x)^2}} dx \right) = \lim_{t \rightarrow e^-} [\sin^{-1}(\ln t)] = \sin^{-1}(1) = \pi/2.$$

Thus the given improper integral is convergent and its value is $\pi/2$.